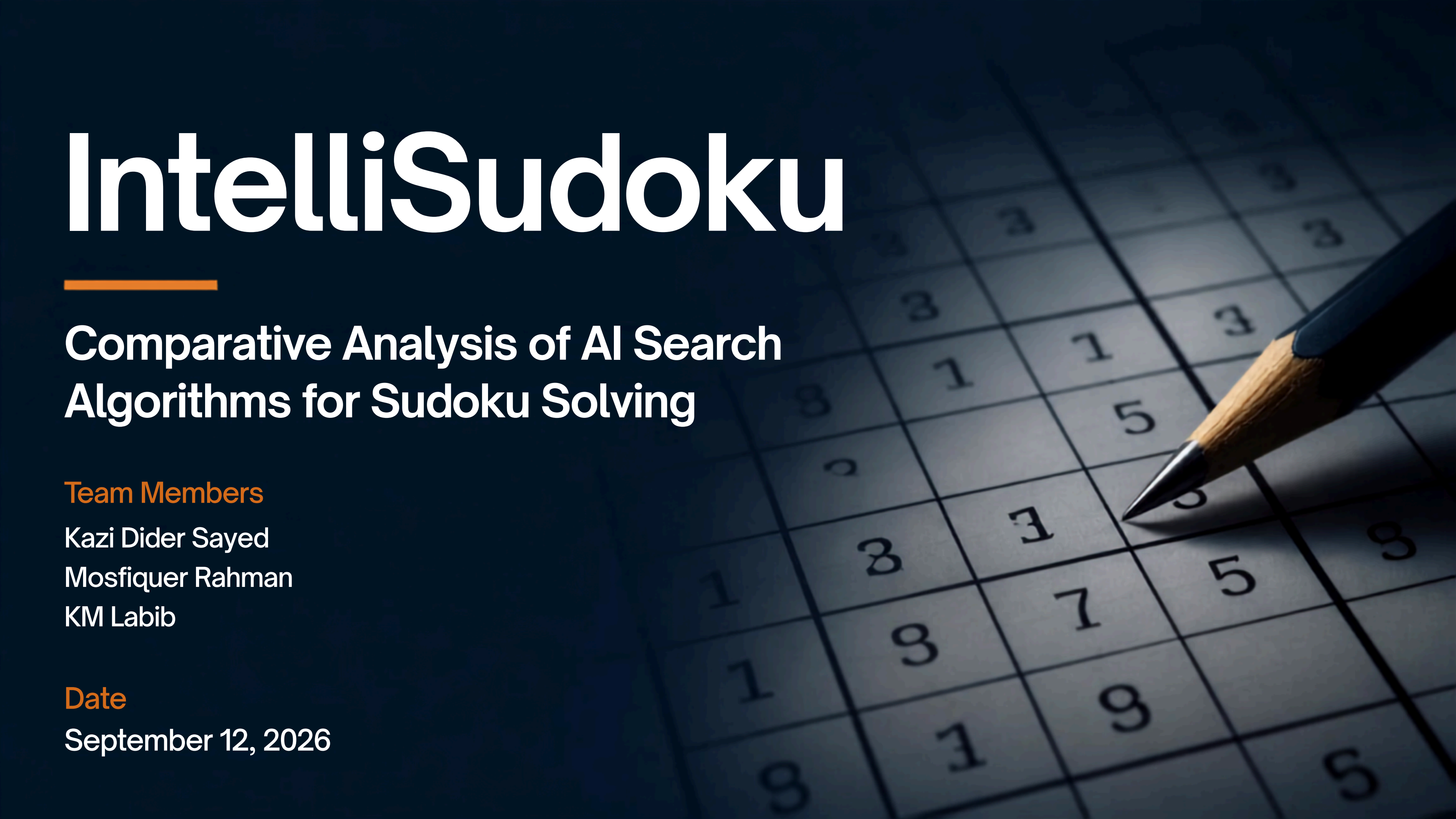


IntelliSudoku



Comparative Analysis of AI Search Algorithms for Sudoku Solving

Team Members

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Problem Statement and Motivation

Sudoku is a constraint-based puzzle that demands extensive search for hard instances

Algorithm choice and heuristics strongly influence solving efficiency

Our goal is to evaluate how heuristics and constraint propagation enhance performance

Focus on puzzles with minimal clues where the challenge is greatest



Dataset Overview

1,000(4×250) Sudoku puzzles
from Kaggle's “1 Million Sudoku Games”
and public 17-clue minimal puzzles

Difficulty levels: **Easy, Medium, Hard, Expert** based on clues and source ratings

Each puzzle includes **grid, difficulty label, verified solution**, and empty cell count

The same puzzles are used for all algorithms to ensure fair comparison



AI Search Algorithms Implemented

01

Basic Backtracking

baseline solver exploring possibilities exhaustively

02

Backtracking + MRV

selects cell with fewer possible values to reduce branching

03

Backtracking + Forward Checking

prunes invalid future paths early

04

MRV + Forward Checking

combines both heuristics for efficiency

05

AC-3 Constraint Propagation

enforces arc-consistency to prune domains before/during search



Evaluation Metrics and Performance Measurement

Solving time

Search nodes explored

Number of backtracks

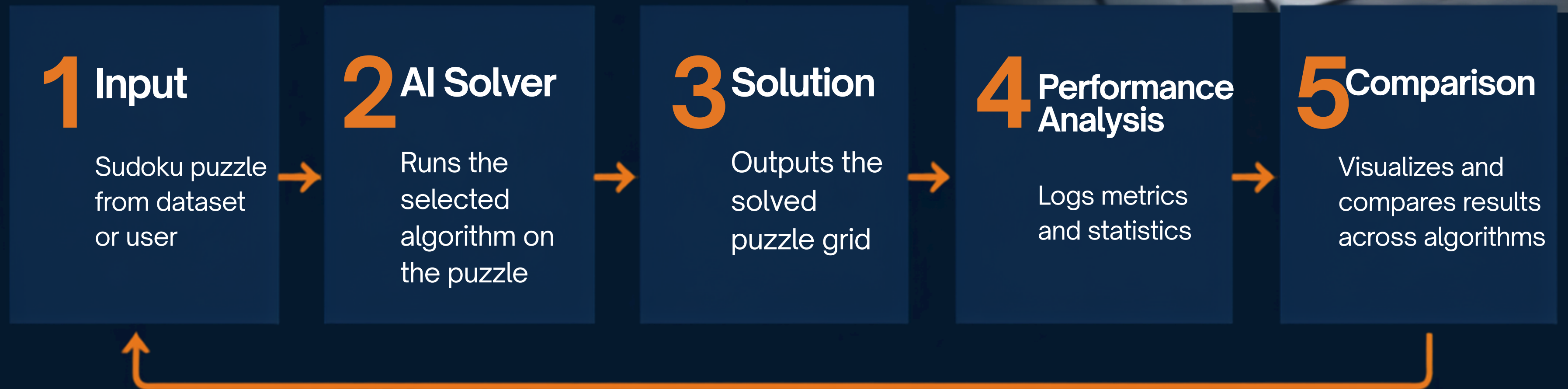
Success rate

Measured consistently across all difficulty levels and algorithms



System Architecture and Workflow

A modular pipeline that solves, analyzes, and compares Sudoku puzzles with AI search algorithms



Expected Outcomes and Closing

Multiple AI Sudoku solvers including AC-3

Cleaned and labeled Sudoku dataset

Performance comparison with tables and graphs

Interactive Sudoku-solving demonstration system

Final technical report and source code repository

Thank you. Questions?

